Edgar Reyes-Rivera

Cincinnati, Ohio
Edgar@ReyesRivera.dev | ReyesRivera.dev
linkedin.com/in/edgar-reyes-rivera | github.com/EdgarReyesRivera

Education

Miami University Oxford, OH
Bachelor of Science in Computer Engineering *Expected May 2027*
Bachelor of Science in Computer Science *GPA: 3.48*

Skills & Certifications

Languages: C++, C, Python, SystemVerilog, Verilog, JavaScript, Java, C#, SQL, CUDA, MIPS Assembly

Hardware & Embedded: FPGA Development, RTL Design, RISC-V Architecture, AXI4 Bus Protocols, Digital Circuit Design, Quartus, Vivado, ModelSim, I2C, SPI, UART, Arduino

Frameworks & Web: React, Vue.js, Firebase, Astro, Tailwind CSS, .NET Framework

Systems & Infrastructure: Linux, Multi-threading, Virtualization (Proxmox), Docker, Git, GitHub

Certifications: Ohio Southwest Alliance on Semiconductors and Integrated Scalable Manufacturing (O.A.S.I.S)

Experience

RoviSys Aurora, OH
Software Co-op *May 2026 – Present*
- Contribute across two concurrent engineering teams supporting software application development and industrial automation systems integration.
- Develop full-stack features in C#, .NET Framework, and Vue.js, managing databases with SQL and SSMS and deploying with Docker.
- Design control modules, implement ladder logic, and configure interlocks in Studio 5000 for industrial automation systems.
- Create and configure HMI screens in FactoryTalk View to provide operator interfaces for automated manufacturing processes.

Synchrony Cincinnati, OH
Software Engineer Intern *July 2024 – September 2024*
- Engineered and delivered technical solutions on a remote, agile Enablement Architecture team, contributing to B2B software for loan processing and private-label credit cards.
- Researched and prototyped software packages, providing data-driven recommendations to meet team requirements.
- Utilized Atlassian tools (JIRA, Bitbucket, Confluence) to manage project tasks and collaborate on code development.

Projects

TinyRV1 RISC-V Processor reyesrivera.dev/projects/tinyrv1-processor
- Designed and implemented a single-cycle, non-pipelined 32-bit RISC-V processor in Verilog on a DE1-SoC FPGA, executing the eight-instruction TinyRV1 subset (arithmetic, multiply, loads/stores, jumps, branch).
- Built a two-pass Python assembler covering all five RISC-V encoding formats to generate the processor's memory images.
- Verified the design with self-checking ModelSim testbenches and four on-board test programs; added push-button single-stepping with live register inspection on 7-segment displays.

Trading Strategy Research System reyesrivera.dev/projects/trading-research
- Developed a Python trading-strategy research system: a SQLite data pipeline covering 25 tickers of daily bars since 2010, a backtest engine, and a 66-test suite.
- Engineered the backtester for trustworthy results using next-bar-open fills, slippage modeling, and an anti-lookahead assertion harness that re-runs strategies on truncated history to prove no future data leaks.
- Optimized the research loop so a full 75-combination strategy sweep completes in about 15 seconds; runs live against the Alpaca paper-trading API since July 2026.

Focus & Habit Dashboard reyesrivera.dev/projects/habit-tracker
- Built a habit, task, and focus-timer dashboard as a single static HTML file using React 18 and Tailwind CSS with no build step; deployed and in daily use at habits.reyesrivera.dev.
- Implemented an offline-first architecture: localStorage persistence with Google sign-in, real-time Cloud Firestore sync across devices, and automatic first-login data migration.
- Designed a drift-free count-up focus timer from timestamp arithmetic, with earned break time banked at 20% of completed work.

Leadership & Service

President, Society of Hispanic Professional Engineers (SHPE) *August 2025 – May 2026*
- Lead student chapter, delegate roles to the executive board, and represent SHPE at university-wide meetings.
- Establish and maintain collaborations with corporate sponsors and other campus organizations.

Relevant Coursework

Embedded Systems Design, Digital Systems Design, Computer Organization, Signals and Systems, Algorithms I, Data Abstractions & Structures, Systems I & II, Database Systems